

Superstore Marketing Campaign Analysis

Customer segmentation + response prediction + Power BI dashboard

The business problem is campaign targeting

The goal is to improve campaign efficiency by finding customers most likely to accept membership offers.

14.91%

Overall response rate

Only 334 of 2,240 customers accepted the campaign, so broad outreach wastes budget.

Key decision

Which customer groups should marketing prioritize for the next campaign?

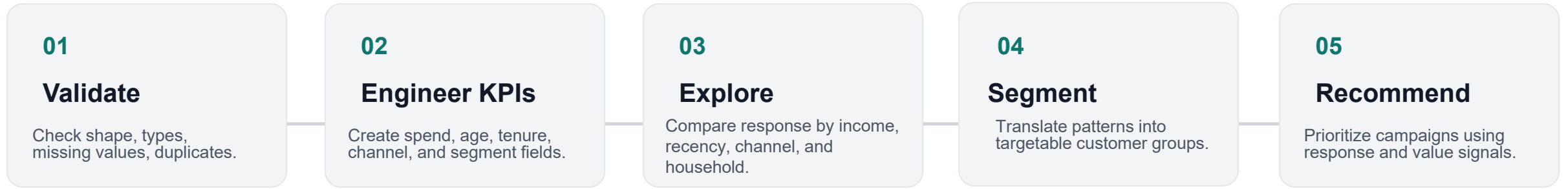
KPI frame: response rate, customer count, average spend, high-value share.

Business questions

Who responds?
What behaviors matter?
Which segments deserve budget?
How should the dashboard guide action?

The workflow turns raw data into decisions

The notebook and dashboard follow a business-first path instead of isolated Pandas practice.



Feature engineering used for decisions

Total spend, total purchases, discount purchase rate, spend per purchase, customer tenure, age group, income bracket, recency group, preferred channel, high-value flag, and customer segment.

Three signals explain the campaign opportunity

Response is strongest when customer value, recent engagement, and preferred buying channel line up.

Meaning for the business

The next campaign should not treat every customer equally. The data points toward a smaller, higher-probability audience where response likelihood compounds.

26.61%

Very high income

Customers in the top income bracket respond at nearly 1.8x the overall average.

23.90%

Recent customers

Customers with lower recency are much more likely to accept the campaign.

35.62%

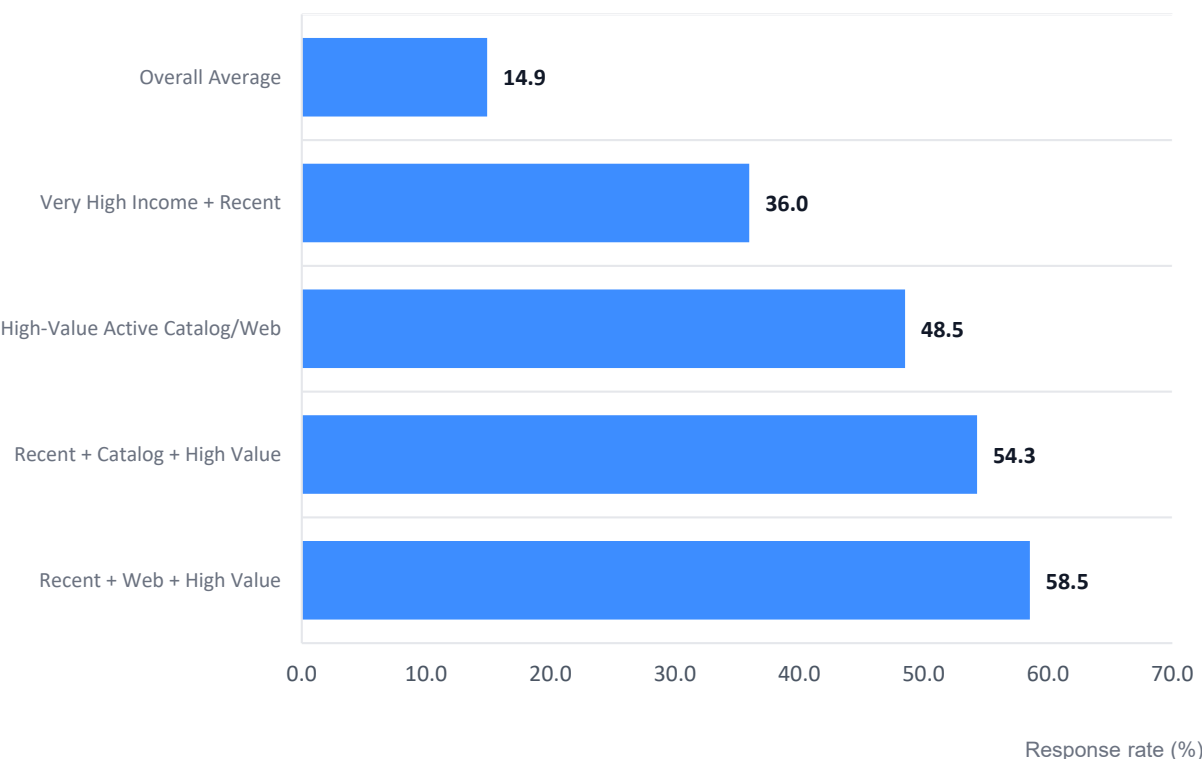
Catalog shoppers

Catalog is the strongest purchase channel signal for campaign acceptance.

Combined signals reveal the strongest target group

Recent high-value Web and Catalog customers respond at more than three times the overall average.

Response rate by target audience



Best audience

58.54%

Recent + Web + High Value

This segment is small, but it is the cleanest starting point for a high-efficiency test campaign.

Baseline response: 14.91%

Segments translate analysis into campaign actions

Each segment ties customer behavior to a practical next step for marketing.

Segment	Customers	Response	Avg Spend	Campaign action
High-Value Active Catalog/Web	134	48.51%	1,479.61	Prioritize premium offers
Catalog Shoppers	50	22.00%	679.04	Catalog-led follow-up
General Customers	291	21.31%	1,025.71	Personalized test campaign
High-Value At-Risk	224	20.54%	1,498.65	Win-back offer
Digital Shoppers	488	17.21%	356.93	Web/email promotion
Bargain Hunters	505	8.71%	121.00	Low-cost offer only

48.51%

Top segment response

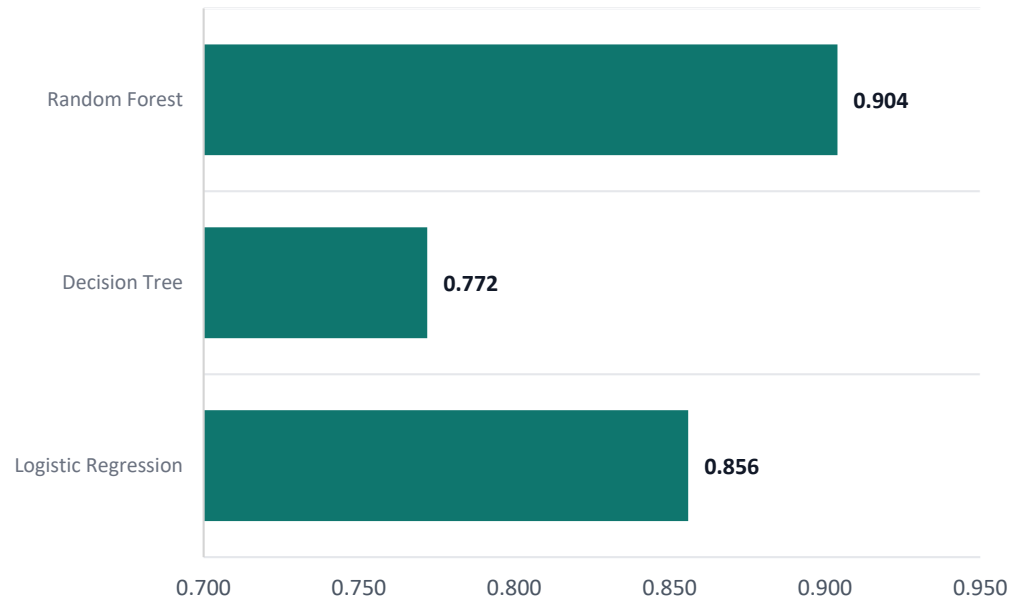
1,479.61

Avg spend in top segment

Use the segment table as a campaign priority list: protect budget for high-response, high-spend groups first.

The model supports targeted outreach

Because the response class is imbalanced, ROC-AUC, recall, precision, and F1 matter more than accuracy alone.



Best model

Random Forest

Accuracy 0.879
Precision 0.623
Recall 0.493
F1-score 0.550
ROC-AUC 0.904

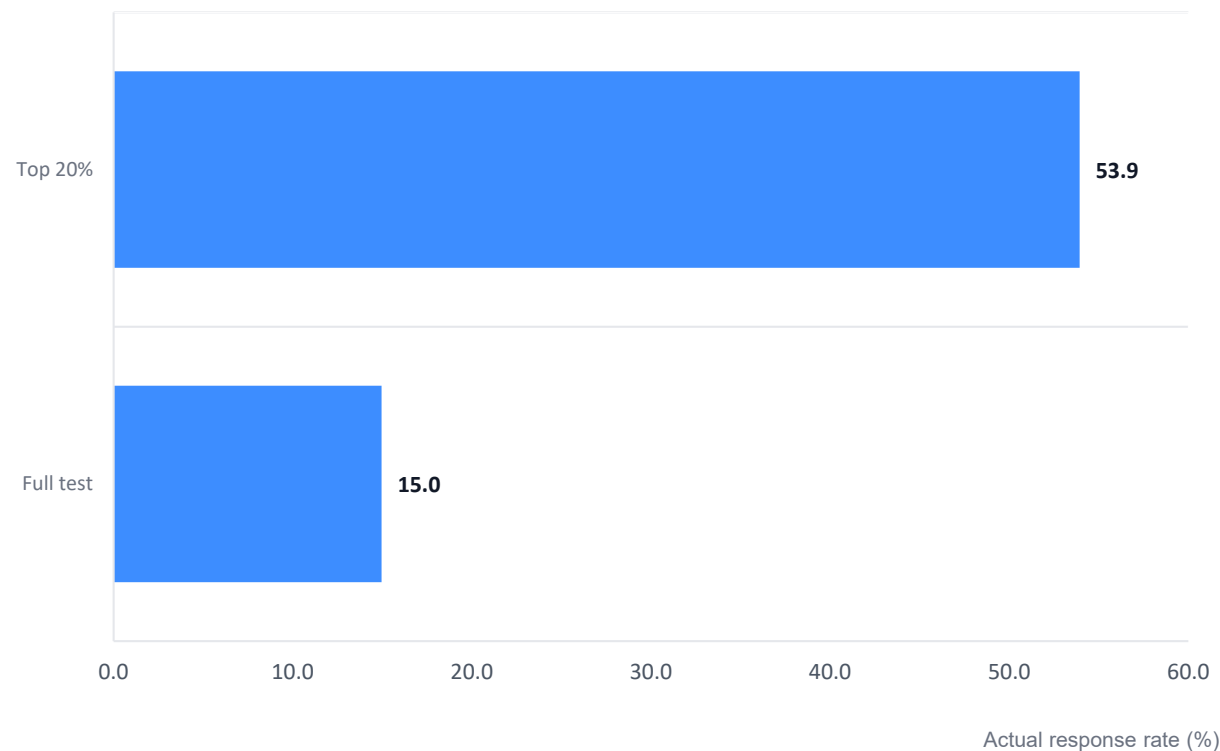
Interpretation

The model is useful for ranking customers by likelihood, then sending targeted campaigns to the highest-scoring groups.

Model scoring lifts response in the top 20%

Ranking unseen test customers by predicted probability concentrates actual responders.

Actual response rate by targeting approach



Targeting lift

3.61x

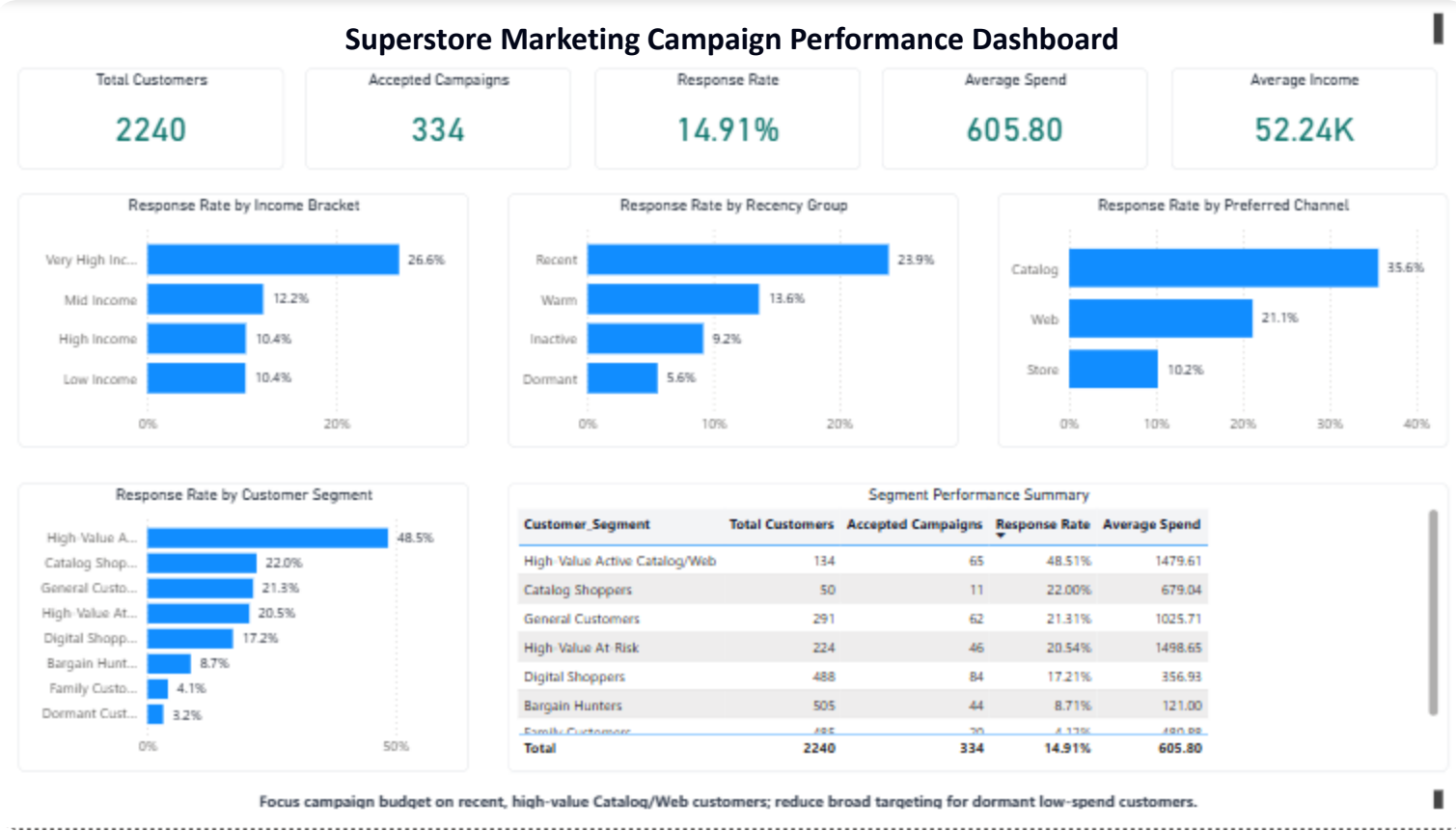
Top 20% predicted customers

The model selected 89 of 448 test customers and captured a much higher actual response rate.

Full test baseline: 14.96%

The dashboard makes the recommendation actionable

Power BI turns the notebook findings into an executive view for filtering, explaining, and prioritizing campaign audiences.



Dashboard view: campaign KPIs, response drivers, and segment performance summary.

Recommendation: focus spend where response compounds

The business should move from broad outreach to segmented campaign execution.

01 **Prioritize recent high-value Catalog/Web customers**

Start with the highest response audience for premium offers and personalized follow-up.

02 **Use focused win-back offers for high-value at-risk customers**

Do not ignore high spenders with longer recency; use urgency, loyalty, or bundle offers.

03 **Limit spend on dormant and low-spend customers**

Use low-cost channels or test-only budgets until these groups show stronger intent.

Expected impact: higher conversion per campaign spend, lower wasted outreach, and a clear audience for the next A/B test.